



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences

First Year - Semester II Examination - August/September 2024

MAT 1202 - NUMERICAL ANALYSIS I

Time: Two (02) hours.

Answer ALL (04) questions.

1. a) Let $a = 0.54617$ and $b = 0.54601$. Use four-digit arithmetic to approximate $a - b$ and calculate the relative errors using i) rounding and ii) chopping.

(45 marks)

- b) Let $f(x) = x - 2e^{-x}$.

- i. Prove that the equation $f(x) = 0$ has a root r between 0 and 1.
- ii. Apply the Bisection method to approximate r . Perform five iterations starting with the interval $(0, 1)$. Given that the exact value of r is 0.85260, calculate the absolute error after the fifth iteration.

(55 marks)

2. State the Newton-Raphson and Secant formulas for finding the n th approximation of a simple real root of the equation $f(x) = 0$.

(10 marks)

- a) Consider the equation: $x^3 + 5x + 1 = 0$. Use the Newton-Raphson method to approximate the real root of the equation. Perform three iterations, starting with an initial guess $x_0 = 0$.

(40 marks)

Contd.

- b) A root r of the equation $xe^x - 1 = 0$ lies in the interval $(0.5, 1.0)$.

Using the secant method, estimate r by performing three iterations.

(50 marks)

3. a) The error function $\text{erf}(x)$ is defined by the integral:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Approximate $\text{erf}(0.08)$ by using Lagrange linear interpolating polynomial in the given table of correctly rounded values.

x	0.05	0.10	0.15	0.20
$\text{erf}(x)$	0.05637	0.11246	0.16800	0.22270

(45 marks)

- b) Consider the following data table:

x	0.5	1.5	3.0	5.0	6.5	8.0
$f(x)$	1.625	5.875	31.0	131.0	282.125	521.0

- Construct divided difference table for the above data.
- Find the Newton's interpolating polynomial, and hence approximate the value of the function f at $x = 7$.

(55 marks)

4. a) Using the Taylor theorem, derive the following central difference formula:

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2),$$

where $h > 0$.

Consider the following data:

x	0.6	0.7	0.8	0.9	1.0
$f(x)$	5.2296	3.1655	2.7531	2.2771	2

Approximate the second derivative $f''(x)$ at the point $x = 0.8$ using the derived central difference formula.

(40 marks)

- b) The value of π can be calculated from the integral:

$$\pi = \frac{1}{2} \int_{-1}^1 \frac{4}{1+x^2} dx.$$

Approximate π using the composite Simpson's 1/3 method with 10 sub-intervals.

(60 marks)

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